

# MongoDB Detailed Notes & Important Questions

## 1. Introduction to MongoDB

- MongoDB is a NoSQL, document-oriented database.
- Data is stored in collections and documents instead of tables and rows.
- Documents are stored in BSON (Binary JSON) format.
- MongoDB is highly scalable, flexible, and suitable for modern web applications.

## 2. Creating a Database in MongoDB

- A database is a container for collections.
- The 'use dbName' command creates or switches to a database.
- The database is physically created when data is inserted into it.

## 3. Embedded Documents in MongoDB

- Embedded documents are documents stored inside another document.
- They help represent one-to-one and one-to-many relationships.
- Embedding improves read performance and reduces joins.

## 4. CRUD Operations in MongoDB

- CRUD stands for Create, Read, Update, and Delete.
- Create: `insertOne()`, `insertMany()`.
- Read: `find()`, `findOne()`.
- Update: `updateOne()`, `updateMany()`.
- Delete: `deleteOne()`, `deleteMany()`.

## 5. Find vs FindOne

- `find()` returns multiple documents as a cursor.
- `findOne()` returns only the first matching document.
- `find()` is used when multiple results are required.

## 6. Inserting Documents

- Documents are inserted using `insertOne()` and `insertMany()`.
- Each document contains key-value pairs.
- MongoDB automatically creates an `_id` field if one is not provided.

## 7. Updating Documents

- `updateOne()` updates a single document.
- `updateMany()` updates multiple documents.
- Common update operators include `$set`, `$inc`, and `$unset`.

## 8. Deleting Documents

- `deleteOne()` removes a single document.
- `deleteMany()` removes multiple documents.
- `drop()` removes an entire collection.

## 9. Projection (Selecting Columns)

- Projection is used to include or exclude specific fields.
- It improves performance by returning only the required data.
- The `_id` field is returned by default unless explicitly excluded.

## 10. Deleting a Database

- The `dropDatabase()` method deletes the current database.
- All collections and documents inside the database are permanently removed.

## 11. Schema Validation

- Schema validation enforces rules on documents.
- It ensures that only valid data is inserted into collections.
- Validation can check data types, required fields, and value ranges.

## 12. MongoDB Atlas (Managed vs Unmanaged Databases)

- Managed databases are hosted and maintained by cloud providers like MongoDB Atlas.
- Unmanaged databases require manual setup and maintenance.
- MongoDB Atlas provides backups, monitoring, scalability, and security features.

# Most Important Exam Questions

1. What is MongoDB? Explain its features and advantages.
2. Differentiate between SQL and MongoDB.
3. What are collections and documents in MongoDB?
4. How do you create and delete a database in MongoDB?
5. Explain Embedded Documents with an example.
6. Explain CRUD operations in MongoDB.
7. Differentiate between find() and findOne().
8. How do you insert, update, and delete documents?
9. What is Projection in MongoDB?
10. What is Schema Validation and why is it important?
11. Differentiate between Managed and Unmanaged databases.
12. Explain MongoDB Atlas and its advantages.